

Static Characteristics of Electronic Instruments

Accuracy • Precision • Sensitivity •
Linearity • Drift • Hysteresis

Definition

- Static characteristics describe instrument performance for steady or slowly varying inputs
- Applicable to DC voltage, temperature, pressure, etc.
- Do not involve time-dependent response

Accuracy

- Closeness of measured value to the true value
- Expressed as % of full scale or % of reading
- High accuracy implies low error

Precision

- Closeness of repeated measurements to each other
- Independent of true value
- Instrument can be precise but inaccurate

Repeatability and Reproducibility

- Repeatability: Same results under identical conditions
- Reproducibility: Similar results under different conditions
- Important for consistency of measurements

Sensitivity

- Ratio of change in output to change in input
- Indicates responsiveness of the instrument
- Example: Volts per degree Celsius ($\text{V}/^{\circ}\text{C}$)

Range and Span

- Range: Minimum and maximum measurable values
- Span: Difference between maximum and minimum values
- Defines operating limits of instrument

Linearity

- Degree to which output follows a straight-line relationship
- Deviation causes non-linear error
- Better linearity improves accuracy

Resolution

- Smallest detectable change in input
- Depends on instrument scale or digital bits
- Higher resolution gives finer measurements

Drift

- Slow change in output with time for constant input
- Caused by temperature changes or aging
- Types: Zero drift and Sensitivity drift

Hysteresis

- Different outputs for same input during increasing and decreasing cycles
- Caused by friction, magnetic effects, or mechanical play
- Leads to measurement error

Dead Zone, Error and Tolerance

- Dead zone: Input range with no output response
- Static error: Difference between true and measured value
- Tolerance: Permissible deviation from specified value

Repeatability

- Ability of a measurement or process to produce consistent results
- Repeated measurements under exactly the same conditions:
- Same observer, same instrument, same equipment
- Same location, same method, short time interval
- Indicates precision and stability of the instrument/system